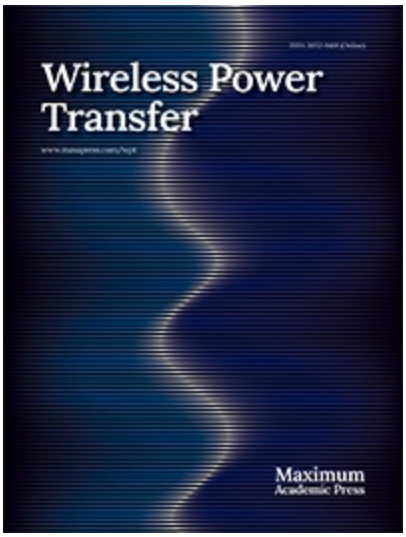


A magnetic coupling structure with dual D coils featuring high anti-offset properties is used for wireless charging of underwater vehicles

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Title Page

A Cylindrical Double-D Magnetic Coupler With High Misalignment Tolerance for AUV Wireless Charging

Running title: Cylindrical DD Coupler for AUV Wireless Charging

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Abstract

Misalignment during underwater docking can cause coupling variations and output instability in wireless charging systems for autonomous underwater vehicles (AUVs). To address this issue, a cylindrical double-D (DD) magnetic coupler integrated with a dual-sided LCC compensation network is proposed. Both the transmitter and receiver consist of two series-connected DD coils conformally arranged along the cylindrical AUV hull. The circumferentially symmetric magnetic field and complementary flux paths improve the tolerance to lateral and full-angle rotational misalignments, while the dual-sided LCC network mitigates the influence of mutual-inductance variations on the reflected impedance and resonant condition. The developed system operates at 85 kHz with input and output voltages of 24 V, a rated output current of 4.4 A, and a rated output power of 105.6 W. Simulation results show that the coupling coefficient varies from 0.6009 to 0.605 over a full circumferential rotation from 0° to 360°, corresponding to a variation of only 0.0041. No evident change in the primary H-bridge midpoint-voltage waveform or system resonant condition is observed under a lateral misalignment of 50 mm or a rotational misalignment of 180°. Prototype experiments further demonstrate stable operation and a regulated output current of 4.4 A under lateral misalignment of 5 and 10 mm, lateral misalignment of 90° and 180°, and light-, rated-, and heavy-load conditions. These results verify the combined advantages of the proposed coupler in lateral misalignment tolerance, omnidirectional rotational adaptability, and load robustness, providing a practical solution for reliable autonomous underwater charging of AUVs subject to docking errors and attitude disturbances.

Key words: underwater vehicle; wireless power transmission; Cylindrical double-D coil; Magnetic coupling mechanism; Double LCC Compensation; Misalignment tolerance

Introduction

The ocean covers approximately 71% of the Earth's surface area and harbors strategic resources such as minerals, biological assets, and energy. It serves as a critical domain for ocean observation, submarine engineering operations and maintenance, and national defense security monitoring. Autonomous Underwater Vehicles (AUVs), free from cable constraints, rely on onboard energy sources, navigation control, and sensors to achieve fully autonomous underwater operations. Characterized by a wide operational range, strong concealment, and flexible deployment, AUVs have become a key development direction in the field of underwater equipment. Although AUVs continue to advance in terms of hull materials, navigation algorithms, and sensor integration, their four core performance metrics—endurance, cruising speed, operational radius, and payload capacity—remain severely limited by onboard energy storage and replenishment systems[1-5].

At present, AUVs generally utilize lithium battery packs as their core power supply. Limited by energy density, a single charge can only sustain continuous operation for a few hours to a few days. The necessity to frequently return to a mother ship for battery replacement or wired charging significantly reduces operational efficiency and elevates mission costs, failing to meet the demands of novel missions such as long-term deep-sea residency and large-scale networked exploration. As

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AUVs continuously evolve towards informatization and high integration, the shortcoming in energy supply has become a critical bottleneck restricting their performance enhancement and application expansion[6]. Under the practical condition that the energy density of power batteries is unlikely to achieve a short-term breakthrough, enhancing AUV endurance has become a research hotspot in ocean engineering. Current mainstream electrical energy replenishment methods have obvious defects: manual surfacing and recovery for charging incurs high costs and cannot support long-duration, far-sea missions; contact charging utilizing underwater cables with wet-mateable connectors is prone to safety hazards such as current leakage and short circuits due to the high conductivity of seawater. Therefore, marine resource exploration and development urgently require new underwater energy replenishment technologies that are safe, reliable, and highly automated. Wireless Power Transfer (WPT) technology, with its advantages of high safety, high automation, strong environmental adaptability, and excellent concealment, can effectively solve the underwater power replenishment dilemma for AUVs, providing a crucial technical pathway to breakthrough endurance bottlenecks[7].

Misalignment tolerance and power-transfer stability of magnetic couplers remain major concerns in AUV underwater WPT systems. A conical-coil AUV wireless charging system was proposed[8]. Compared with conventional planar coils used in air, the conical coils provided improved tolerance to flow-induced positional disturbances and maintained a relatively stable coupling coefficient within a limited misalignment range. The system achieved an output power of 500 W with a power-transfer efficiency exceeding 90%. A bell-mouth-shaped magnetic coupler was subsequently developed, increasing the maximum transfer power to 1000 W while maintaining an efficiency above 80%[9]. However, neither configuration fully addressed the lateral and angular misalignments encountered during practical AUV docking, resulting in relatively high sensitivity to multidirectional positional deviations.

To reduce the sensitivity of magnetic couplers to positional deviations, nested annular coupling structures were proposed[10]、[11].Based on an operating principle similar to that of an autotransformer, these structures consist of two nested inner and outer solenoidal coils. They can mitigate the influence of rotational misalignment on power-transfer performance while improving the coupling coefficient, thereby facilitating high-power and high-efficiency power transfer. Nevertheless, the annular geometry restricts autonomous AUV entry into the charging station and increases the required installation space, thereby reducing the convenience and reliability of underwater docking.

Considering the curved AUV hull and the positional disturbances induced by ocean currents, an E-shaped magnetic-core coupler with a certain degree of misalignment tolerance was introduced[12].Under open-loop experimental conditions, the system achieved a transfer power of 120 W with an efficiency exceeding 95%. An I-shaped magnetic-core coupler was presented to reduce receiver weight while maintaining compatibility with the curved AUV hull[13]. The corresponding system achieved a maximum output power of 600 W and a power-transfer efficiency

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of 88.3%. However, the coupling performance of the E- and I-shaped configurations degrades under lateral, longitudinal, and roll misalignments. Conventional annular couplers exhibit strong rotational-misalignment tolerance but occupy considerable installation space inside the AUV and may expose onboard electronic equipment to the charging magnetic field, thereby compromising electromagnetic compatibility and usable layout space. Segmented annular magnetic-core structures, including annular couplers equipped with dual and triple transmitting coils, were proposed[14–17]. These configurations were designed to maintain approximately constant mutual inductance during relative rotation between the transmitter and receiver, thereby reducing power-transfer fluctuations caused by AUV roll misalignment. [18]-[19]

To improve the misalignment tolerance of AUV underwater wireless charging systems, this paper proposes a cylindrical double-D (DD) magnetic coupler in which both the transmitter and receiver consist of two series-connected DD-coil units. The complementary magnetic-flux paths and circumferentially symmetric arrangement reduce variations in mutual inductance under lateral and rotational misalignments. A dual-sided LCC compensation network is subsequently designed to mitigate the influence of coupling variations on the reflected impedance and resonant condition. The coupler parameters are optimized through finite-element analysis, and the system performance is evaluated through electromagnetic–circuit co-simulation and prototype experiments.

Coupling System Modeling and Cylindrical Double-D Mechanism Design

Characteristic Analysis of the LCC Compensation Network

In a wireless charging system, the compensation network is the critical link determining transmission efficiency and output stability. Under misalignment conditions, the magnetic coupling structure will cause severe fluctuations in mutual inductance and reflected impedance, subsequently leading to instability in the system's output current and voltage, as well as a significant drop in transmission efficiency. Therefore, selecting a compensation network with stable output characteristics that is insensitive to variations in coupling parameters is key to ensuring the continuous and reliable operation of the AUV wireless charging system under misalignment conditions.

After comprehensively comparing various compensation topologies, the LCC compensation network was selected for the system in this paper due to its strong adaptability and ease of implementation. The LCC compensation network achieves resonance by connecting capacitors in series at both the transmitting and receiving ends, offering the dual advantages of high efficiency, simple structure, and good stability. Its resonant state remains unaffected by load and mutual inductance.

The equivalent circuit diagram of the LCC compensation network is shown in Figure 1. Here, L_P and L_S represent the self-inductance values of the primary and secondary coils, respectively; C_{P1}、C_{P2}、C_{S1} and C_{S2} represent the primary and secondary coil compensation capacitors; L_{P1} and L_{S3} are the primary and secondary compensation inductors; M denotes the mutual inductance between

the transmitting and receiving coils; R_L is the equivalent load; k is the coupling coefficient; u_{AB} is the input voltage; and u_{ab} is the output voltage.

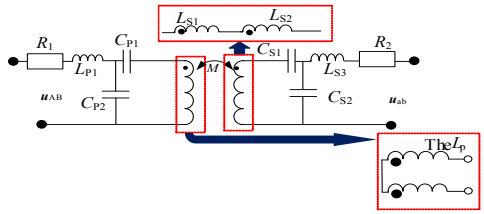


Figure 1 LCC resonant network

At this time, the impedance Z of the system is:

$$Z_{in} = j\omega L_{P1} + \frac{1}{j\omega C_{P1}} + \frac{1}{j\omega C_3 + \frac{1}{j\omega L_P}} \tag{1}$$

The core of the resonant state is to eliminate the inductive and capacitive reactance in the circuit, allowing the total impedance of the system to present a purely resistive characteristic to achieve highly efficient transmission with zero reactive power loss. By making the reactances of the primary and secondary series branches cancel each other out, that is, when the system operates at the resonant frequency ω_0, then:

$$\frac{1}{j\omega C_{P2} + \frac{1}{j\omega L_P}} = \frac{j\omega L_P}{1 - \omega^2 L_P C_{P2}} \tag{2}$$

$$1 - \omega^2 L_P C_{P2} = 0 \tag{3}$$

At this point, the impedance of the parallel branch approaches infinity, and all reactive components on the primary side are completely canceled out, leaving only the equivalent series resistance R_1. Thus:

$$Z_1 = R_1 \tag{4}$$

The primary-side resonant current I_1 and the secondary-side resonant current I_2 are respectively:

$$I_1 = \frac{\boldsymbol{u}_{AB}(R_2 + R_L)}{R_1(R_2 + R_L) + \omega^2 M^2} \tag{5}$$

$$I_2 = \frac{j\omega M \boldsymbol{u}_{AB}}{R_1(R_2 + R_L) + \omega^2 M^2} \tag{6}$$

$$u_{ab} = \frac{j\omega R_L}{R_2 + R_L} I_1 \tag{7}$$

The system output power P_{out} is the power consumed by the secondary current on the equivalent load, i.e.:

$$P_{out} = I^2 R_L \approx \frac{\boldsymbol{u}_{AB} R_L}{(\omega_0 M)^2} \tag{8}$$

$$P_{out} = I^2 R_L \approx \frac{\boldsymbol{u}_{AB} R_L}{\left(\omega_0 k \sqrt{L_P L_S}\right)^2} \tag{9}$$

When R_1 and R_2 are relatively small, as the double-D magnetic coupling mechanism undergoes misalignment or rotation along with the AUV, the mutual inductance M and the coupling

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coefficient k will experience dynamic changes. However, the resonant state of the LCC network is determined solely by L_P, C_P, and L_S, C_S, remaining unaffected by fluctuations in M. This ensures the system can maintain high-efficiency resonant transmission across all operating conditions, creating a synergy with the structure's anti-misalignment capabilities. The output power is directly proportional to the square of the coupling coefficient k^2. The mutual inductance (M) and coupling coefficient (k) vary when the proposed cylindrical DD magnetic coupler is subjected to positional misalignment, resulting in a corresponding variation in the reflected impedance. However, under the designed compensation conditions, the resonant frequency of the dual-sided LCC network is primarily determined by the compensation inductors and capacitors rather than by the mutual inductance. Consequently, the network can maintain a stable resonant condition and reduce the influence of coupling variations on the inverter input impedance and system output.

Structural Characteristics and Transmission Principle of the DD-type Magnetic Coupling Mechanism

The DD-type magnetic coupling mechanism consists of two symmetrical D-shaped coils connected in series in the same direction. It possesses advantages such as concentrated magnetic fields, high coupling strength, low magnetic leakage, and strong misalignment tolerance. Compared to traditional single coils, its magnetic field superposition effect is more significant, and the magnetic flux path is more regular. It can maintain stable coupling under lateral, longitudinal, and rotational misalignments, effectively suppressing performance degradation caused by dislocation. Concurrently, this structure exhibits strong magnetic field confinement capabilities, which can reduce stray interference and enhance electromagnetic compatibility, making it highly suitable for AUV underwater wireless charging scenarios.

The power transmission of the DD coil follows the basic principle of high-frequency electromagnetic induction coupling. When high-frequency sinusoidal alternating current is applied to the DD coil on the transmitting side, an alternating coupling magnetic field is excited in space. The DD coil on the receiving side cuts the magnetic flux within this alternating magnetic field, generating an induced electromotive force that supplies power to the load after passing through the compensation network and rectification/filtering. Assuming the self-inductances of the two sets of D coils on the transmitting side are L_{P1} and L_{P2}, respectively, with a total self-inductance of L_P, and the mutual inductance between the coils is M_{DD1}, then the total self-inductance at the transmitting end is:

$$L_P = L_{P1} + L_{P2} + 2M_{DD1} \tag{10}$$

Similarly, the total inductance L_S on the receiving side is:

$$L_S = L_{S1} + L_{S2} + 2M_{DD2} \tag{11}$$

In Equation (8), L_{S1} and L_{S2} are the self-inductances of the two sets of D coils at the receiving end, and M_{DD2} is the mutual inductance inside the receiving coils. The equivalent mutual inductance Meq between the transmitting and receiving ends is formed by the superposition of the coupling mutual inductances among the four sets of coils:

$$M_{eq} = M_{11} + M_{12} + M_{21} + M_{22} \tag{12}$$

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From the equations, it is evident that the DD coil can significantly enhance transmission capacity by improving the mutual inductance M_{eq} and the coupling coefficient k. Simultaneously, it stabilizes M_{eq} under misalignment conditions, thereby ensuring a smooth output.

The AUV underwater wireless charging system is susceptible to ocean current disturbances and docking deviations. Misalignment of the magnetic coupling mechanism will cause a sharp decrease in the mutual inductance of traditional single coils and fluctuations in reflected impedance, resulting in a significant decline in transmission efficiency and output stability. The DD-type magnetic coupling mechanism, relying on its symmetrical dual-coil structure and magnetic field complementary effect, possesses excellent misalignment tolerance characteristics: the two sets of D coils form redundant magnetic flux paths. When the coupling on one side weakens, the other side can maintain effective linkage, keeping the equivalent mutual inductance stable. To quantitatively evaluate the misalignment tolerance of the DD coil, the mutual inductance retention rate ξ is defined as:

$$\xi = \frac{|M_{mis}|}{M_0} \times 100\% \tag{13}$$

Where M_0 is the equivalent mutual inductance without misalignment, and M_{mis} is the equivalent mutual inductance after misalignment. A higher mutual inductance retention rate indicates stronger coupling stability under misalignment. The variation in the coupling coefficient Δk can be expressed as:

$$\Delta k = \frac{|M_{mis} - M_0|}{\sqrt{L_P L_S}} \tag{14}$$

The smaller Δk is, the more superior the system's misalignment tolerance performance. Furthermore, misalignment will directly alter the system's reflected impedance Z_{ref}, the expression for which is:

$$Z_{ref} = \frac{\omega^2 M_{eq}^2}{R_L + R_S} \tag{15}$$

The DD coil can maintain M_{eq} at a high degree of stability during the misalignment process, thereby suppressing severe jumps in the reflected impedance Z_{ref}, reducing reactive power fluctuations, and keeping the input impedance angle stable. The reduced variation in the reflected impedance helps stabilize the input-impedance angle and provides favorable conditions for zero-voltage switching (ZVS) of the inverter,reducing switching losses and EMI interference.

Design of the Cylindrical Double-D Magnetic Coupling Mechanism

Driven by the core requirement to enhance the system's misalignment tolerance against ocean current disturbances and docking deviations in AUV wireless charging systems, a cylindrical double-D (DD) magnetic coupling mechanism is proposed. Its structure is shown in Figure 2. Both the transmitting and receiving ends of this structure utilize two sets of DD coils connected in series. The coils are connected end-to-end along the circumferential direction and arranged symmetrically to form a closed cylindrical morphology. This configuration can maintain stable magnetic coupling in the radial, axial, and 360° circumferential rotational directions, effectively rectifying the

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inadequate coupling performance of traditional planar coils and planar DD coils under misalignment conditions.

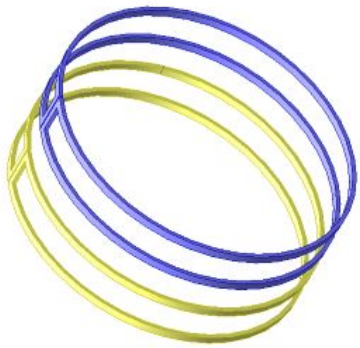


Figure 2 Cylindrical double-D-shaped magnetic coupling structure

The analysis results of the magnetic field directions generated by the transmitting coils are shown in Figure 3 and Figure 4. The proposed cylindrical DD magnetic coupler is formed by conformally bending two DD-coil units along the cylindrical surface and arranging them symmetrically in the circumferential direction. Both the transmitting and receiving assemblies contain two series-connected DD-coil units. Each DD-coil unit consists of two adjacent D-shaped windings connected such that their currents generate a concentrated transverse magnetic field. This conformal arrangement preserves the magnetic-field characteristics of the planar DD coil while adapting the coupler to the curved AUV hull.

The current directions and corresponding magnetic-field distributions are illustrated in Figs. 3 and 4. In the cross-sectional view shown in Fig. 3, conductor segments A and B carry currents that generate magnetic fields in the same direction, while segments C and D generate fields in the opposite direction. These conductor groups correspond to paths 1–2 and 3–4 in Fig. 4, respectively. The superposition of their magnetic fields establishes a closed radial magnetic-flux path between the two DD-coil groups. The main flux is therefore concentrated in the power-transfer region, while the external leakage field is reduced.

The two symmetrically arranged DD-coil groups provide complementary magnetic-flux paths. When lateral misalignment weakens the flux linkage associated with one coil group, the other group retains sufficient flux linkage, thereby reducing variations in the equivalent mutual inductance and coupling coefficient. During circumferential rotation, the cylindrical symmetry maintains an approximately unchanged magnetic-path length, reluctance, and relative coil geometry. This configuration therefore provides both lateral-misalignment tolerance and full-angle rotational adaptability without relying on the previously used DL- or DCC-coil analogies.

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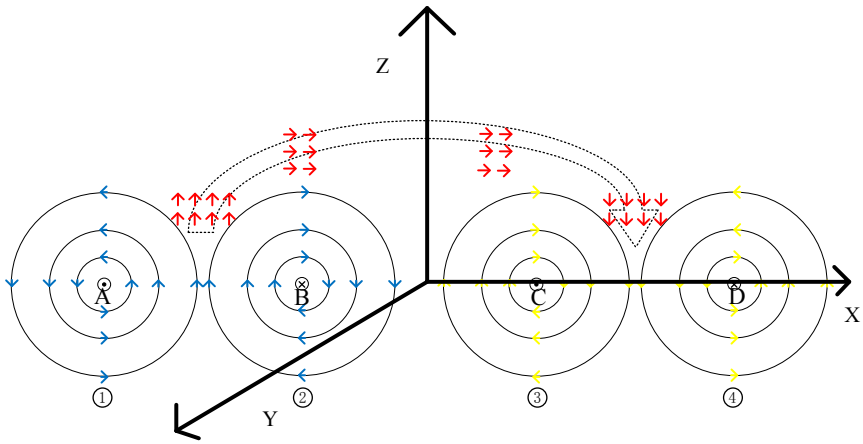


Figure 3 A cross-sectional profile of the magnetic field generated by the current within a cylindrical DD coil

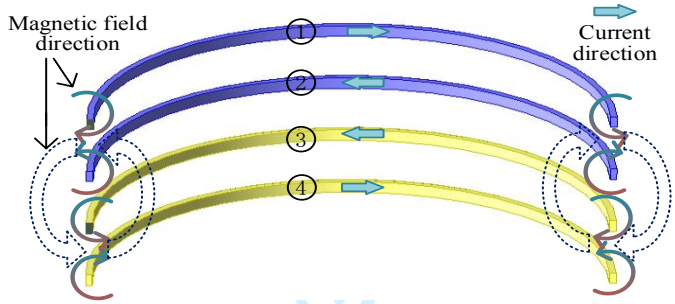


Figure 4 Schematic diagram of the magnetic field space of the cylindrical DD coil

As can be seen from Figure 3 and Figure 4, the cylindrical double-D magnetic coupling mechanism achieves high misalignment tolerance relying on circumferentially symmetrical magnetic field distribution and the principle of magnetic flux complementary coupling. The DD coils on the transmitting side are arranged symmetrically along the circumference of the cylinder. Once energized, they can excite a closed, uniform, and directionally constrained annular coupling magnetic field in space. The main magnetic flux efficiently links along the radial direction of the cylinder, and the leakage flux components are significantly suppressed. When lateral or longitudinal misalignment occurs, the coupling magnetic flux of one set of D coils weakens, while the other set can still maintain sufficient magnetic flux linkage. Through magnetic field superposition and redundant compensation, the overall equivalent mutual inductance and coupling coefficient maintain a smooth transition, avoiding the sharp drop in mutual inductance and sudden plunge in efficiency typical of traditional single coils. When the receiving end undergoes a 360° rotational misalignment, the axisymmetric characteristic of the cylindrical structure ensures that the magnetic path length, magnetic reluctance, and magnetic flux distribution remain fundamentally unchanged. The coupling characteristics will not experience periodic abrupt changes with angular deflection, mechanistically guaranteeing the stability of transmission performance under rotational conditions.

Simulation Analysis

For the cylindrical DD-type magnetic structure, the receiving coil has the same shape as the transmitting coil. The receiving end can be installed conformally with the outer hull of the AUV,

while the transmitting end is fixed to the underwater charging base. The overall structure is shown in Figure 5, where the AUV radius is 200 mm, and the transmitting coil radius is 240 mm.

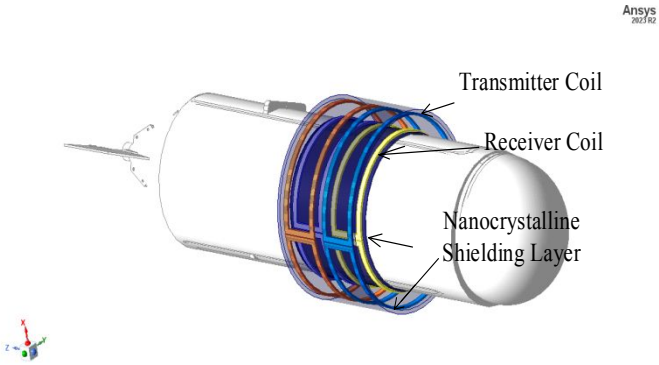


Figure 5 Schematic diagram of the wireless charging structure for the Double-D type AUV

The proposed coupler combines the concentrated transverse magnetic field of the DD-coil units with the rotational invariance provided by the circumferentially symmetric cylindrical arrangement. The DD-coil units concentrate the main magnetic flux in the power-transfer region, whereas the cylindrical symmetry limits variations in the relative coil geometry and magnetic path during rotation. To evaluate these characteristics, the rotational angle and lateral position of the receiving coupler were varied under different transmitting-coil dimensions. The resulting relationships between the coupling coefficient, rotational angle, and lateral misalignment are presented in Figure 6 and Figure 7, respectively.

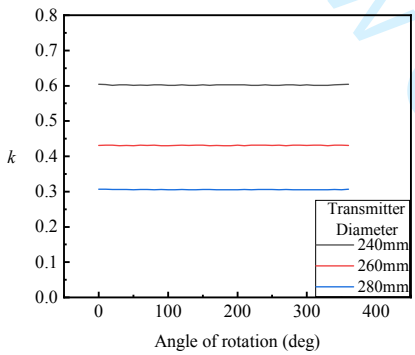


Figure 6 The relationship between the coupling coefficient of the cylindrical DD-type magnetic structure and the rotation angle under different transmission distances

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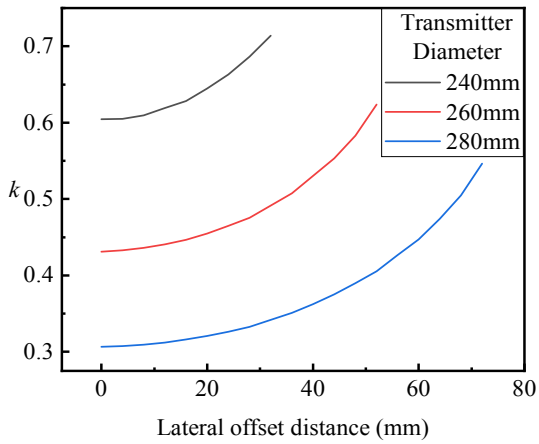


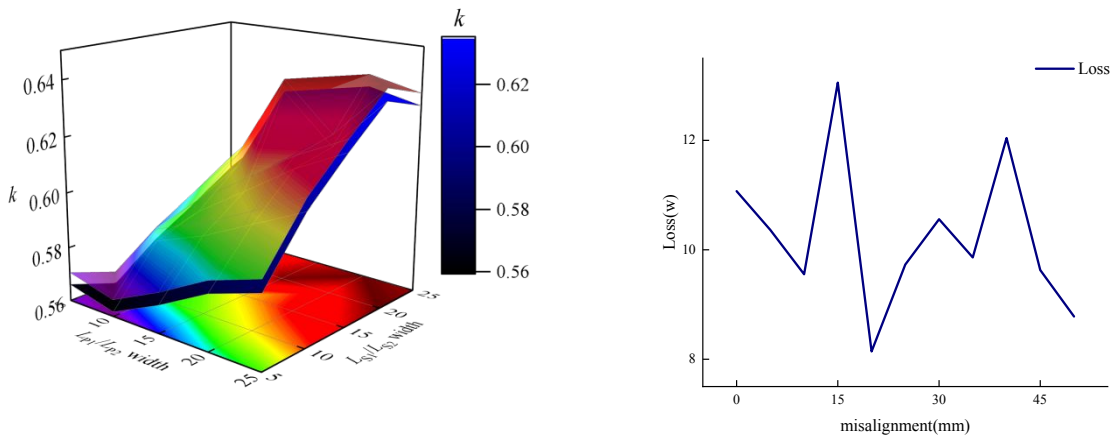
Figure 7 The variation relationship of the coupling coefficient of the cylindrical DD-type magnetic structure with the lateral misalignment distance under different transmission distances

To verify the anti-rotational and anti-lateral misalignment performance of the cylindrical DD magnetic coupling mechanism, this study conducted a swept simulation of the receiving end's rotational angle and lateral displacement under different transmitting coil radius parameters. The variation rules of the coupling coefficient with the rotational angle and lateral misalignment distance were obtained, with results shown in Figure 6 and Figure 7, respectively.

As seen in Figure 6, when the transmitting coil radius is 240 mm, the receiving end undergoes a full circumferential rotational misalignment of 0°~360°. The maximum coupling coefficient is 0.605, and the minimum is 0.6009, with an amplitude of fluctuation not exceeding 0.01, which is far below the average coupling coefficient level. Moreover, the trend curves under different transmitting coil radii are basically identical, indicating no distinct correlation between coupling coefficient fluctuation and transmitting coil size. This demonstrates that the magnetic structure can still maintain a highly stable magnetic coupling state under full-angle rotational conditions. It possesses excellent omnidirectional rotational misalignment resistance, effectively addressing the random attitude deflection issues during AUV underwater docking.

To clarify the impact of coil parameters on the system's coupling characteristics, a sweeping analysis of the widths of the transmitting and receiving coils was performed to explore their individual and synergistic mechanisms on the coupling coefficient. Keeping parameters such as coil turns and transmission distance constant, the radial widths of the transmitting and receiving coils were altered respectively. A surface relationship of the coupling coefficient with varying coil widths was obtained through finite element simulation.

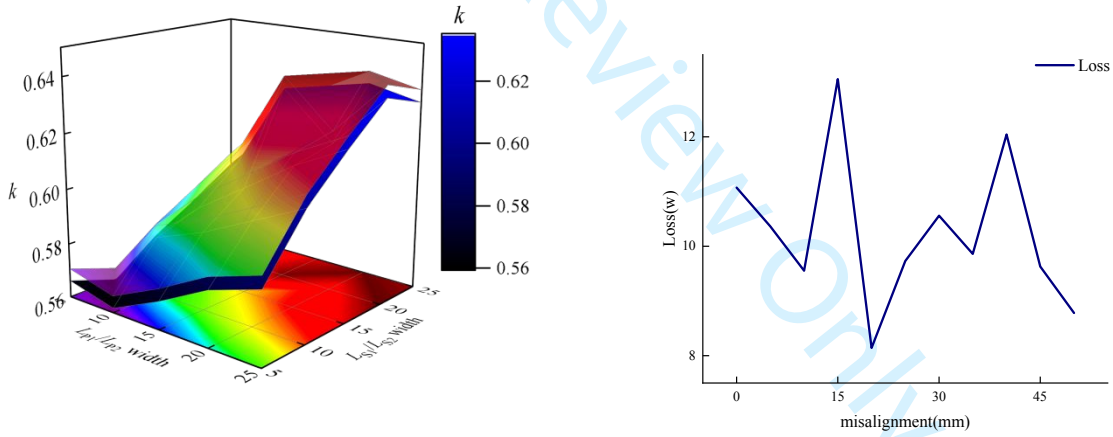
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(a) (b)

Figure 8 (a)the influence of the width of the transmitting coil and the receiving coil on the coupling coefficient(b) Power Loss Versus Misalignment

As shown in



(a) (b)

Figure 8 (a)the coupling coefficient k exhibits a monotonically increasing trend and tends to saturate as the transmitting coil width L_P increases. When L_P gradually increases from a smaller value, the effective coupling area of the coil significantly expands, and the main magnetic flux linkage rapidly rises, with k quickly ascending from approximately 0.56. When L_P exceeds a certain critical value, magnetic field edge effects and leakage losses begin to manifest, and the growth rate of the coupling coefficient notably slows down, entering a plateau phase. The impact of the receiving coil width L_S on the coupling coefficient demonstrates a similar variation to L_P, but it is more sensitive regarding its influence on the surface morphology. Under identical L_P conditions, increasing L_S can significantly enhance magnetic flux capture capability, further elevating k. However, when L_S is excessively large, the parasitic parameters of the coil itself increase, and the coupling gain

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gradually flattens out. The highest point of the surface appears in a parameter combination area where both L_P and L_S are relatively large and matched to each other; at this time, the coupling coefficient can reach approximately 0.64, representing the optimal coupling state for this structure. If the difference in width between the two is too large, the surface will show an obvious depression, and the coupling coefficient will drop significantly, indicating that the degree of matching between the widths of the transmitting and receiving coils is a key factor determining magnetic coupling efficiency. Therefore, comprehensively considering the maximization of the coupling coefficient and the constraints of AUV installation space, a parameter combination where the transmitting and receiving coil widths are both 25 mm and the coil radius is 240 mm is selected as the optimized configuration. Subsequent comparative analysis of misalignment characteristics is carried out on this basis.

To quantify the effect of lateral misalignment on system loss in a conductive seawater environment, a lateral-misalignment sweep was performed using the seawater-medium model under identical excitation conditions, as shown in Fig. 8(b). Over the investigated misalignment range of 0–50 mm, the simulated system loss fluctuates between approximately 8 and 13 W and does not exhibit a sustained increase with increasing misalignment. This variation is mainly attributed to changes in mutual inductance, reflected impedance, and leakage-field distribution, together with the redistribution of induced eddy currents in the conductive seawater. The loss remains within a relatively limited range, indicating that the proposed coupler does not experience continuous loss deterioration under the investigated seawater and misalignment conditions.

Further analysis of Figure 7 reveals that under lateral misalignment conditions in the X-Y plane, the coupling coefficient exhibits a consistent variation pattern with the misalignment distance: the coupling coefficient is at its minimum when the receiving end is located at the center of the transmitting mechanism. As the lateral misalignment amount gradually increases, the coupling coefficient continues to rise with a significant amplitude. Addressing this characteristic, auxiliary positioning structures such as mechanical guides, gravity self-centering, or truncated cone positioning can be added to the underwater charging base, or transmitting coils with a smaller diameter can be selected. This can effectively restrict the misalignment range of the AUV receiving end, ensuring the system consistently operates in the high coupling coefficient region and guaranteeing an efficient and stable charging process. To verify the magnetic field characteristics of the cylindrical DD-type magnetic structure, simulation software was employed to analyze the model shown in Figure 5 ultimately yielding its cross-sectional magnetic field distribution, as shown in Figure 9

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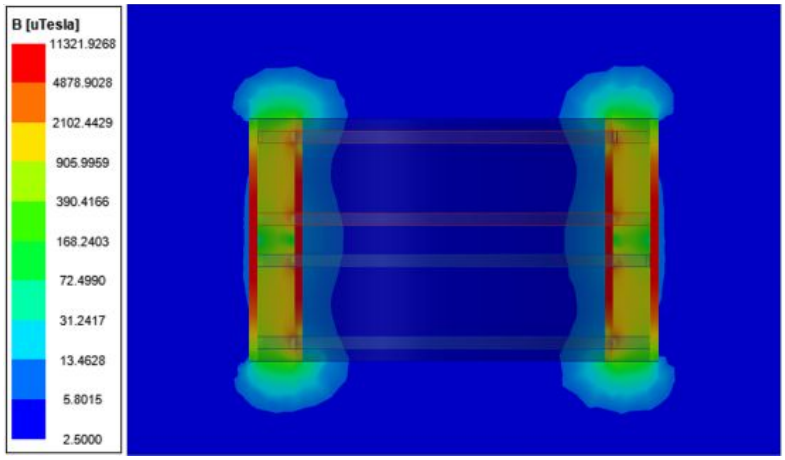


Figure 9 The spatial magnetic field distribution of the cylindrical DD-type magnetic structure

As shown in Figure 9, the cylindrical DD-type magnetic structure can effectively concentrate the magnetic field between the transmitting coil and the receiving coil. This structural design ensures that the magnetic field strength inside the receiving coil and outside the transmitting coil remains relatively low. This helps reduce the interference caused by seawater and metal materials on the magnetic field in marine environments and scenarios containing diverse metallic substances, thereby reducing system power loss and improving energy transmission efficiency.

As shown in Fig. 9, the magnetic field in the seawater environment is primarily concentrated in the power-transfer region between the transmitter and receiver, whereas the field strength inside the receiving coil and outside the transmitting coil remains relatively low. This field-confinement characteristic reduces the time-varying magnetic flux linked with the surrounding conductive seawater and metallic AUV hull and explains, from the perspective of magnetic-field distribution, why the system loss in Fig. 8(b) does not continuously increase with misalignment. From the mechanical-integration perspective, the proposed cylindrical DD coupler is considered technically feasible. The receiver can be conformally mounted on the outer surface of the cylindrical AUV hull, while the transmitter is fixed to the underwater charging station. This arrangement does not occupy the internal space of the AUV pressure hull and does not require the AUV to pass through a closed annular coil, thereby avoiding structural obstruction during autonomous docking. The two DD-coil sets adopt a symmetric segmented arrangement and can be integrated as modular components using curved nonmagnetic supports, waterproof dielectric encapsulation, and a streamlined protective enclosure. The optimized coil dimensions are compatible with the AUV hull curvature and charging clearance. Moreover, the scaled three-dimensional-printed prototype has been assembled and tested under aligned and misaligned conditions, confirming the suitability of the curved-surface installation, available installation space, and assembly path. Therefore, the proposed coupler provides not only strong electromagnetic misalignment tolerance but also practical mechanical manufacturability, installability, and docking compatibility. Deep-water pressure, seawater corrosion, and long-term thermal cycling can be further addressed through

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pressure-resistant encapsulation, corrosion-resistant materials, and heat-transfer paths to the AUV hull.

Simulation Experiment Analysis

To meet the application requirements of high-efficiency transmission, high robustness, and misalignment tolerance for the AUV underwater wireless charging system, this section adopts a dual LCC series-parallel hybrid compensation topology. This topology possesses constant voltage output characteristics, inverter soft-switching operational capability, and strong resistance to parameter deviation, effectively matching the cylindrical double-DD magnetic coupling mechanism. Based on Ansys Maxwell 3D and Simplorer, a field-circuit co-simulation system is constructed to completely realize the synergistic simulation of the magnetic coupling mechanism and the compensation network, as shown in Figure 10.

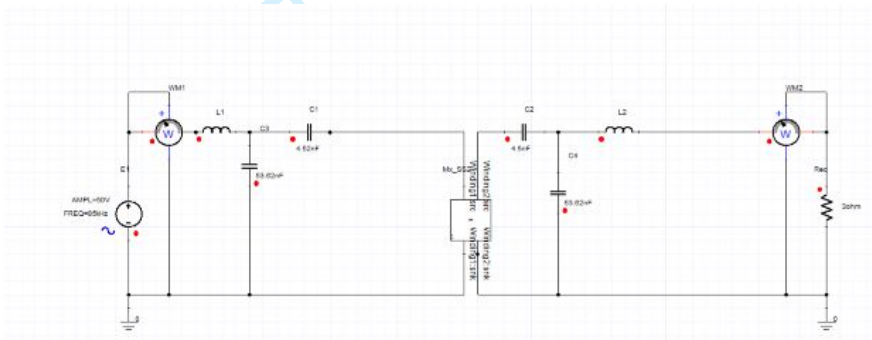


Figure 10 Joint Simulation Platform

An AUV wireless charging magnetic coupling mechanism model is established in Maxwell 3D, with the structure shown in Figure 5. The AUV radius is 200 mm, the transmitting coil radius is 240 mm, and the coil width is 25 mm. Both the transmitting and receiving coils are made of Litz wire, and the magnetic core utilizes a high-permeability nanocrystalline material. Winding excitation, mesh refinement, and eddy current field solving are configured to complete the extraction of self-inductance and mutual inductance parameters, generating a Reduced-Order Model (ROM) for the Simplorer circuit co-simulation. The ROM generated by Maxwell 3D is imported into Simplorer. First, the dual LCC compensation topology is built, and values are assigned according to the simulated parameters. Port mapping among the primary winding, secondary winding, and circuit network is completed. Input voltage, load, simulation duration, and observation probes are set before launching the field-circuit co-transient simulation.

Based on the electromagnetic-circuit co-simulation platform built by Ansys Maxwell 3D and Simplorer, a baseline condition simulation with zero misalignment is first conducted on the cylindrical DD magnetic coupling mechanism and the dual LCC compensation system. The midpoint voltage waveform of the primary H-bridge is obtained as a performance baseline. Building upon this, two typical misalignment conditions—a 50 mm radial misalignment and a 180° rotational misalignment—are simulated. The corresponding primary H-bridge midpoint voltage waveforms are extracted and subjected to comparative analysis.

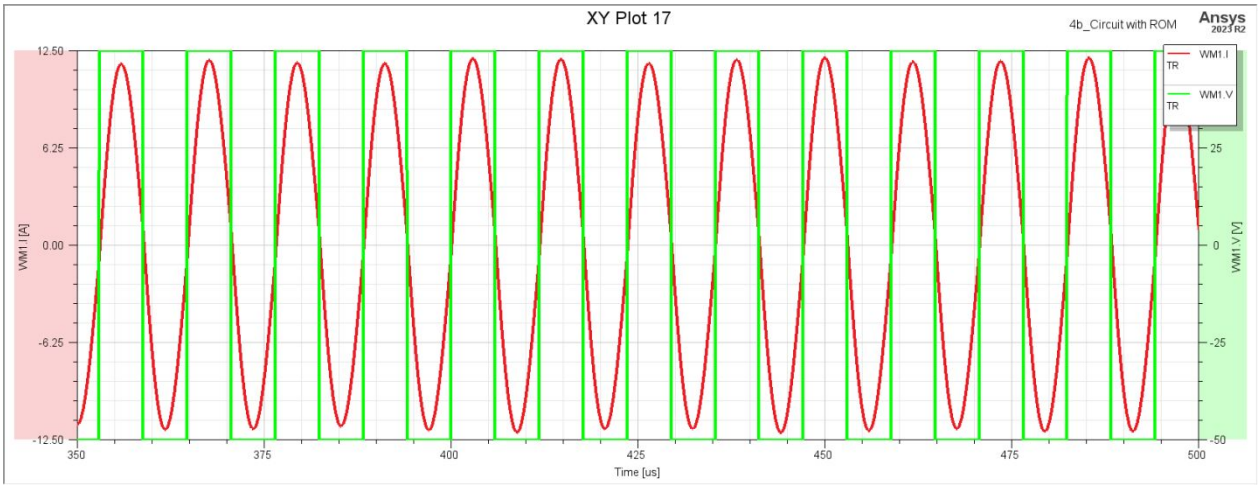


Figure 11 The midpoint voltage of the H-bridge in normal operating condition

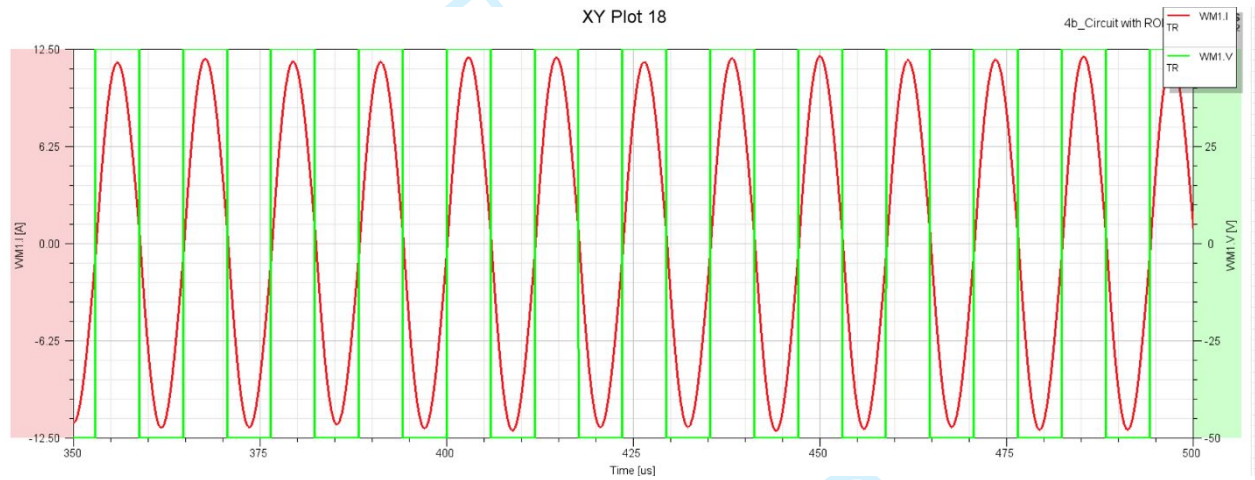


Figure 12 The midpoint voltage of the H bridge in the radial misalignment of 50 mm state

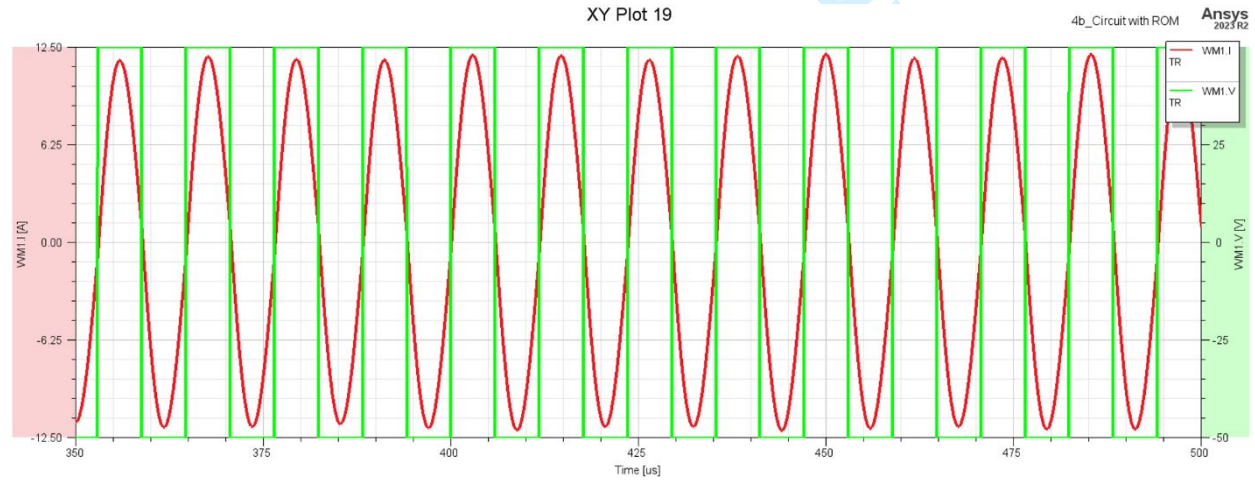


Figure 13 The midpoint voltage of the H-bridge in the state of 180° rotation misalignment

Comparing the waveforms under the three working conditions, it is evident that under the 50 mm radial misalignment and 180° rotational misalignment, the frequency, amplitude, and square wave morphology of the primary H-bridge midpoint voltage do not undergo significant changes, showing only negligible minor perturbations at the edges. This result demonstrates that the designed cylindrical DD magnetic coupling mechanism and the dual LCC compensation system

possess excellent misalignment robustness. Under misalignment conditions, they can still maintain stable resonant states and electrical characteristics, ensuring the inverter continuously operates within the soft-switching region. This provides strong support for the reliable operation of the AUV underwater wireless charging system in scenarios involving docking deviations and attitude disturbances.

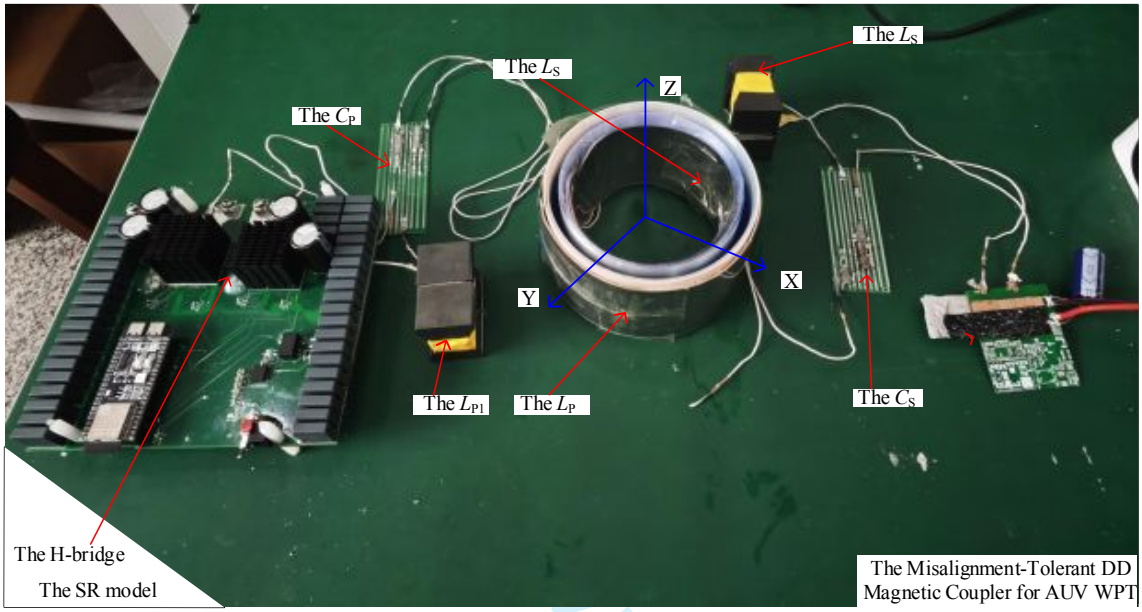


Fig14. Experimental prototype

To validate the effectiveness of the proposed AUV wireless charging system, an experimental prototype employing the double-D (DD) magnetic coupler was developed, as shown in Fig. 14. The coupler was designed to provide high tolerance to positional misalignment. On the transmitter side, a high-frequency full-bridge inverter based on TI LMG3522R30 GaN devices was used to excite the resonant network. On the receiver side, INNOSCIENCE INN150LA070A GaN devices and a synchronous rectification controller were employed to implement a full-bridge synchronous rectifier, thereby reducing rectification losses and improving the overall system efficiency. An FTP020-300-16 dc power supply and an FT63212A-600-75 dc electronic load were used to emulate the system dc input and the AUV lithium-battery load, respectively. The dc–dc power-transfer efficiency was calculated from the real-time measurements obtained from the dc power supply and electronic load. To maintain geometric consistency between the experimental prototype and the practical AUV platform, a scaled AUV hull and its wireless charging assembly were fabricated using three-dimensional printing. Both the transmitting and receiving coils were wound using 0.05 mm × 350-strand Litz wire. The main design parameters of the system are listed in Table I. The input and output voltages were both set to 24 V, corresponding to the dc supply voltage and the maximum charging voltage of the 24-V AUV propulsion battery, respectively. Under the fully aligned condition, the measured coupling coefficient was 0.347.

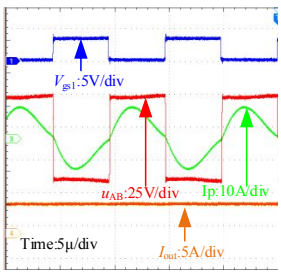
TABLE I

Experimental Parameters

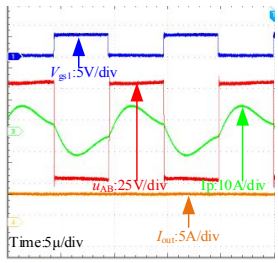
Spec./Par	Val.

Input DC voltage U_{in} /V	24
Output current I_o /A	4.4
Output DC voltage U_{out} /V	24
Maximum power /W	105.6
Resonant frequency f /kHz	85
The transmitting coil L_p /μH	36.1
The transmitting capacitance C_{p1} /nF	388
The transmitting capacitance C_{p2} /nF	129.4
Transmitter-side compensation inductor L_{p1} /μH	9.025
The receiver coil L_s /μH	33
The receiver capacitance C_{s1} /nF	424.9
The receiver capacitance C_{s2} /nF	4×80/10
Transmitter-side compensation inductor L_{s2} /μH	8.25
Coupling coefficient k	0.347
Number of Turns for the L_{p1} and L_{p2}	5×4

With a dc input voltage of 24 V, an output current of 4.4 A, and the magnetic coupler in the fully aligned position, the system operated under the rated-load condition. The corresponding experimental waveforms are shown in Fig. 15(e). The primary current i_p was nearly in phase with the H-bridge output voltage u_{AB}, indicating a quasi-zero-phase-angle (ZPA) operating condition. Accordingly, the input impedance seen by the inverter was approximately resistive, satisfying the fundamental operating requirements of a resonant wireless charging system. Meanwhile, the output current remained stable at 4.4 A, meeting the specified design target.The experimental waveforms under lateral misalignments of 5 and 10 mm are shown in Figs. 15(a) and 15(b), respectively. By adjusting the switching frequency of the gate-drive signal V_{gs1}, the system maintained stable resonant operation and a regulated output current under both misalignment conditions. The waveforms corresponding to rotational misalignments of 90° and 180° are presented in Figs. 15(c) and 15(d), respectively, with no evident distortion observed in the key electrical waveforms. Furthermore, as shown in Figs. 15(e)–(g), the system waveforms exhibited no significant variation under rated-, light-, and heavy-load conditions, demonstrating satisfactory load adaptability. These experimental results confirm that the proposed cylindrical double-D magnetic coupler maintains stable power transfer under lateral misalignment, rotational misalignment, and load variations, thereby validating its strong misalignment tolerance and system-level operating robustness.



(a)



(b)

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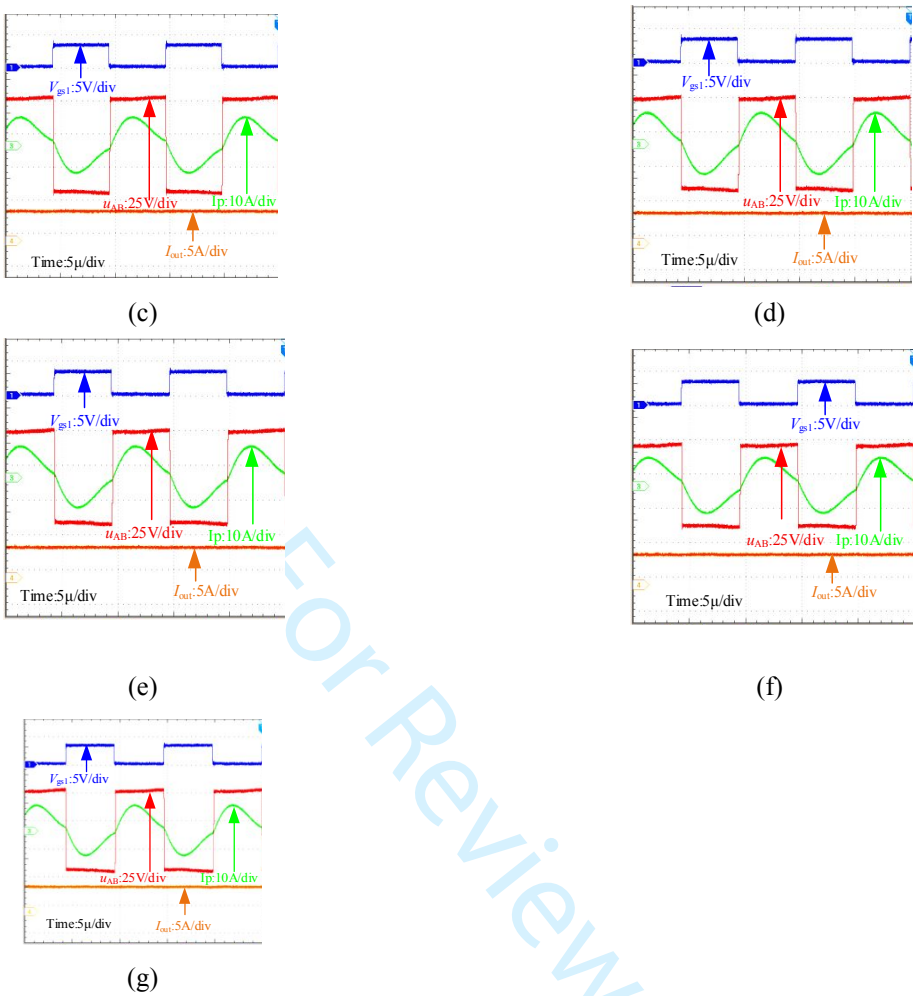


Fig15(a) Waveforms under a 5 mm lateral misalignment(b) Waveforms under a 10 mm lateral misalignment(c) Waveforms under a 90° rotational misalignment(d) Waveforms under a 180° rotational misalignment(e) Waveforms under full-load conditions(f) Waveforms under light-load conditions(g) Waveforms under heavy-load conditions

As summarized in Table II, existing coupler configurations generally exhibit advantages in terms of power level, efficiency, or misalignment tolerance along a specific direction. The reported power levels of the conical- and bell-mouth-shaped couplers reach 500 W and 1000 W, respectively, whereas the E- and I-shaped magnetic-core configurations achieve efficiencies exceeding 95% and 88.3%, respectively. However, these studies primarily focus on aligned operation or limited positional disturbances and do not provide quantitative results for coupling-coefficient variations over a full circumferential rotation from 0° to 360°. Nested annular and segmented annular couplers can improve mutual-inductance stability under roll misalignment; nevertheless, they may require a larger installation volume, restrict access to the docking station, or increase the complexity of multi-coil or multiphase excitation. Because the power ratings, transfer distances, and efficiency evaluation criteria differ among the cited studies, the data in Table II are intended primarily to illustrate the respective performance emphasis of each configuration. Therefore, an entry marked "not reported" does not necessarily indicate that the corresponding coupler lacks that capability.

In contrast, the proposed cylindrical DD coupler is designed considering the geometric constraints of the cylindrical AUV hull and the random attitude deviations encountered during

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underwater docking. Both the transmitter and receiver employ two series-connected DD coils, whose complementary magnetic-flux paths suppress coupling variations caused by lateral and rotational misalignments. Simulation results show that the coupling coefficient varies from 0.6009 to 0.605 over a full circumferential rotation from 0° to 360°, corresponding to (\Delta k=0.0041), which is less than 0.01. The primary H-bridge waveforms also remain stable under a lateral misalignment of 50 mm and a rotational misalignment of 180°. Prototype experiments further verify regulated constant-current output under a lateral misalignment of 10 mm and rotational misalignments of 90° and 180°. Therefore, rather than solely pursuing the highest power level or conversion efficiency, the proposed design addresses the coordinated requirements of full-angle rotational tolerance, lateral-misalignment tolerance, and output stability within a compact cylindrical installation space. This feature makes it particularly suitable for AUV underwater charging applications subject to docking errors and attitude disturbances.

TABLE II

Performance Comparison of Existing AUV Wireless Charging Magnetic Couplers and the Proposed Coupler

Coupler Configuration	Coupling Variation Under Rotational Misalignment	Lateral-Misalignment Tolerance	Power	Main Features and Limitations
Conical Coil [8]	Relatively stable within a limited disturbance range; Δk over 0°–360° not reported	Not quantified; lateral and angular misalignments not fully considered	500W	Tolerates limited flow-induced disturbances; limited tolerance to multi-axis misalignment
Bell-Mouth-Shaped Coupler [9]	Full-circumferential-rotation data not reported	Not quantified; relatively sensitive to lateral and angular misalignments	1000W	High power level; limited adaptability to docking errors
Nested Annular Coupler [10]–[11]]	Mitigates the effects of rotational misalignment; Δk not reported	Not quantified; the annular opening restricts autonomous AUV entry	Not consistently reported	High coupling coefficient and rotational-misalignment tolerance; large installation volume and restricted docking access
E-Shaped Magnetic Core [12]	Coupling performance degrades under roll and other misalignments; Δk not reported	Some tolerance reported but not quantified	120W	High reported efficiency; limited combined tolerance to lateral, longitudinal, and roll misalignments
I-Shaped Magnetic Core [13]	Coupling performance degrades under roll and other misalignments; Δk not reported	Not quantified	600W	Lightweight receiver and compatibility with curved hulls; degraded performance under multidirectional misalignments

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Segmented Annular Coupler [14]–[17]	Designed to maintain approximately constant mutual inductance during rotation; no consistent Δk reported	Focused primarily on roll misalignment; lateral-misalignment data not consistently reported	Varies among studies	Strong rotational-misalignment tolerance; increased structural and multi-coil or multiphase excitation complexity
Proposed Cylindrical DD Coupler	0°–360°: k=0.6009–0.605, Δk=0.0041 (<0.01)	Stable primary H-bridge waveforms at 50 mm in simulation and regulated output at 10 mm experimentally	105.6W	Combines full-angle rotational and lateral misalignment tolerance with cylindrical-hull compatibility; dual-sided LCC compensation maintains stable output

Conclusion

This paper proposes a cylindrical double-D (DD) magnetic coupler combined with a dual-sided LCC compensation network for AUV wireless charging. The complementary magnetic-flux paths and circumferentially symmetric arrangement improve lateral- and rotational-misalignment tolerance. During a complete 0°–360° rotation, the simulated coupling coefficient varies only from 0.6009 to 0.605, corresponding to (Δk=0.0041). In the conductive seawater model, the system loss remains approximately within 8–13 W over a lateral-misalignment range of 0–50 mm. A scaled prototype operating at 85 kHz achieves a rated output of 24 V/4.4 A (105.6 W), with a measured aligned coupling coefficient of 0.347. Stable resonant operation and regulated output are maintained under lateral misalignments of 5 and 10 mm, rotational misalignments of 90° and 180°, and different load conditions. These results verify the misalignment tolerance, output stability, and mechanical integration feasibility of the proposed coupler for AUV underwater charging.

Author contributions

The authors confirm contribution to the paper as follows: study conception and design: Y Hao; draft manuscript preparation: B Wang; manuscript translation: S Yin; data collection and curation: Q Wang, Y Ming; manuscript revision: W Huang. All authors reviewed the results and approved the final version of the manuscript.

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Conflict of interest

The authors declare that they have no conflict of interest.

Data availability

The data that support the findings of this study are available from thecorresponding author upon reasonable request.

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